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COMMON HALF YEARLY EXAMINATION – 2025

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Standard XII

Reg.No.

PHYSICS

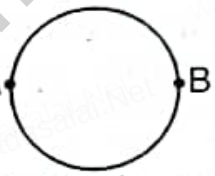
Part – I

Time: 3.00 hrs. Marks: 70

I. Choose the correct answer: 15 x 1 = 15

1. An electric Field $\vec{E} = 10x\hat{i}$ exists in a certain region of space. Then the potential difference $V = V_o - V_A$, where V_o is the potential at the origin and V_A, is the potential at x = 2m is
   a) 10V b) –20V c) + 20V d) –10V
2. Which charge configuration produces a uniform electric field?
   a) point charge b) infinite uniform line charge
   c) uniformly charged infinite plane d) uniformly charged spherical shell
3. Nuclear force is a) electrostatic force b) Long range force
   c) short range force d) magnetic force.
4. In a hydrogen atom, the electron revolving in the fourth orbit, has angular momentum equal to a) h b) h/π c) $4h/\pi$ d) $2h/\pi$
5. A wire of resistance 2 ohms per mater is bent to form a circle of radius 1m. The equivalent resistance between its two diametrically opposite points A and B as shown in the figure is

   a) $\pi\ \Omega$ b) $\frac{\pi}{2}\ \Omega$ c) $2\pi\ \Omega$ d) $\frac{\pi}{4}\ \Omega$



6. The potential energy of magnetic dipole whose dipole moment is $\vec{P_m} = (-0.5\hat{i} + 0.4\hat{j})Am^2$ in uniform magnetic field $\vec{B} = 0.2\hat{i}$ T
   a) –0.1 J b) –0.8 J c) 0.1 J d) 0.8 J
7. $\frac{20}{\pi^2}$ H inductor is connected to a capacitor of capacitance C. The Value of C is order to impart maximum power at 50 Hz is a) 50 μF b) 0.5 μF c) 500 μF d) 5 μF
8. Which of the following is not true for electromagnetic waves?
   a) it transport energy b) it transport momentum
   c) it transport angular momentum
   d) in vacuum, it travels with different speeds which depend on thair Frequency.
9. The gravitational waves were theoretically proposed by a) Conrad Rontgen
   b) Marie Curie c) Albert Einstein d) Edward Purcell
10. The barrier potentiel of a silicon diode is approximately
    a) 0.7 V b) 0.3 V c) 2.0 V d) 2.2 V
11. The principle based on which a solar cell operates is
    a) Diffusion b) Recombination c) Photovoltaic action (d) Carrier flow
12. An object is placed in front of a convex mirror of focal length of F and the maximum and minimum distance of an object from the mirror such that the image formed is real and magnified
    a) 2F and c b) c and ∞ c) f and 0 d) None of these
13. A ray of light travelling in a transparent medium of refractive index 'n' falls on a surface separating the medium from air at an angle of incidents of 45°. The ray can undergo total internal reflection for the following n,
    a) n = 1.25 b) n = 1.33 c) n = 1.4 d) n = 1.5

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14. Light transmitted by Nicol prism is
a) partially polarised b) unpolarised
c) plane polarised d) elliptically polarised
15. In a electron microscope, the electrons are accelerated by a voltage of 14 kV. If the voltage is changed to 224 kV, then the de Broglie wavelength associated with the electrons would
a) increase by 2 times b) decrease by 2 times
c) decrease by 4 times d) increase by 4 times

Part - II

II. Answer any 6 questions. (Q.No.24 is compulsory) **6 x 2 = 12**

16. Define "Electric flux".
17. State the principle of potentiometer.
18. Calculate the radius of $_{79}Au^{197}$ nucleus.
19. What is optical path?
20. What do you mean by Doping?
21. Define work function of a metal.
22. What are the applications of Infrared rays?
23. Write the limitations of Cyclotron.
24. A straight metal wire crosses a magnetic field of flux 4m Wb in a time 0.4s. Find the magnitude of the emf induced in the wire.

Part - III

III. Answer any 6 questions. (Q.No.33 is compulsory) **6 x 3 = 18**

25. Obtain the expression for energy stored in the parallel plate capacitor.
26. State De Morgan's First and Second theorems.
27. Derive the expression for de Broglie wavelength of electrons.
28. Distinguish between drift velocity and mobility.
29. Derive the equation for the effective focal length for lenses in contact.
30. Write down the properties of electromagnetic waves.
31. The wavelength of a light is 450nm. How much phase it will differ for a path of 3 mm?
32. What are the properties of neutrons?
33. A coil of a tangent galvanometer of diametre 0.24 m has 100 turns. If the horizontal component of Earth's magnetic field is 25×10^{-6} T then, calculate the current which gives a deflection of 60°.

Part - IV

IV. Answer all the questions. **5 x 5 = 25**

34. a) Calculate the electric field due to a dipole on its equatorial plane. ***(OR)***
b) Explain the construction and working of a full wave rectifier.
35. a) Discuss the spectral series of hydrogen atom. ***(OR)***
b) Derive the expression for phase angle between the applied voltage and current in a series RLC circuit.
36. a) Obtain a expression for the force on a current carrying conductor placed in a magnetic field. ***(OR)***
b) Explain about compound microscope and obtain the equation for the magnification
37. a) What is absorption spectra? Give their types. ***(OR)***
b) Derive the equation for angle of deviation produced by a prism and thus obtain the equation for refractive index of material of the prism.
38. a) Explain the determination of unknown resistance using meter bridge. ***(OR)***
b) i) Give the applications of photo cell (Any two)
ii) List out the Laws of photoelectric effect (Any three)

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